

Explanation

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1 Provided Repository structure Explanation

I analyzed the local folder. It contains two ROS 2 workspaces:

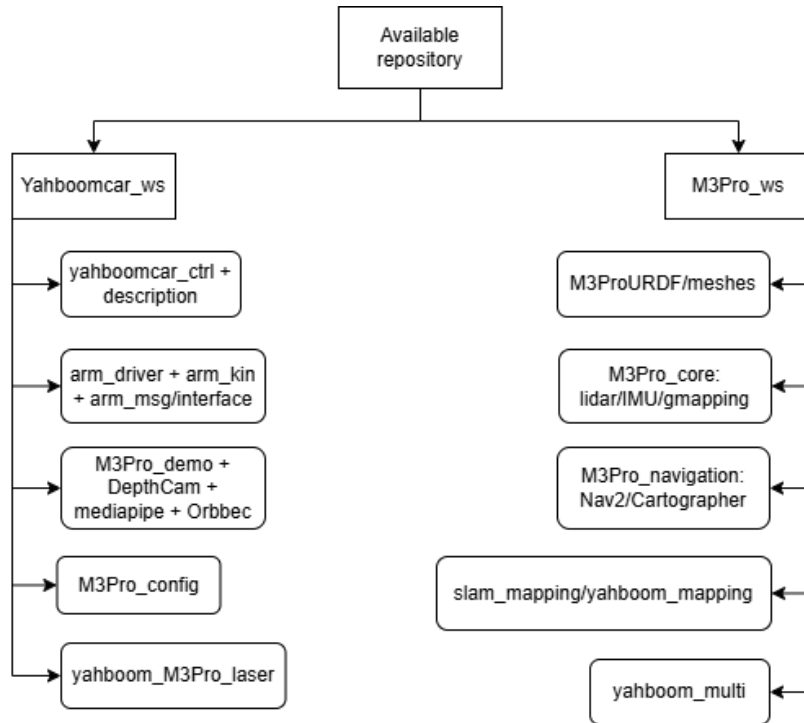


Figure 1: Overview of the downloaded repository structure

1.1 yahboomcar_ws/src

- **yahboomcar_ctrl**: Keyboard and joystick teleoperation. Important files include `yahboom_keyboard.py` and `yahboom_joy_M3Pro.py`.
- **yahboom_M3Pro_description**: Contains the robot URDF, meshes, and RViz model display files.
- **M3Pro_config**: MoveIt2 configuration package containing arm planning settings and fake `ros2_control` configuration.
- **M3Pro_demo**: Includes computer vision and manipulation demonstrations such as AprilTag detection, color tracking, KCF tracking, grasping, and line-following.
- **yahboom_M3Pro_laser**: Lidar-based applications including obstacle avoidance, obstacle warning, and target tracking.
- **yahboom_M3Pro_DepthCam**: Depth camera examples including distance measurement, depth-color visualization, and edge detection.
- **OrbbecSDK_ROS2**: ROS 2 driver package for Orbbec/Dabai depth cameras.
- **arm_interface**, **arm_msgs**: Custom messages and services used for robotic arm communication.
- **arm_kin**: Forward and inverse kinematics services implemented using KDL.
- **arm_driver**: Serial communication driver for the robotic arm using the Rosmaster library.

1.2 M3Pro_ws/src

- **M3Pro**: Robot URDF, meshes, and model display launch files.
- **M3Pro_core**: Core robot packages including laser merging, laser filtering, IMU filtering, and GMapping support.
- **M3Pro_navigation**: Navigation stack configuration including Nav2, Cartographer localization, and RViz navigation setups.
- **slam_mapping**, **yahboom_mapping**: Packages used for SLAM and map generation/saving.
- **ekf_bringup**: Robot localization package that fuses `/odom_raw` and `/imu/data` using an Extended Kalman Filter (EKF).
- **calibration**: Linear and angular calibration utilities based on `/cmd_vel` and TF data.

- **yahboom_multi**: Multi-robot navigation and coordination configurations.
- **patrol**: Laser-based autonomous patrol behavior implementation.
- **largemodel**, **largemodel_arm**: AI, voice interaction, and computer vision application examples.